Cassandra-Risk Paper 3: Cross-Platform Prediction Market Ensembles and Governed Signal Infrastructure

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Abstract

Paper 1 established that a sparse, governed public reconstruction of Cassandra-Risk could generate directionally compelling risk-adjusted results from prediction-market event probabilities. Paper 2 expanded that event universe, documented the resulting degradation, and recovered a coherent architecture consisting of monetary calibration, targeted hazard pruning, and structural concentration caps. This paper, Paper 3, bridges the gap between theoretical risk overlay and governed live signal infrastructure. We evaluate cross-platform ensembles using structurally independent CFTC-regulated prediction markets, finding that Kalshi's tighter Becker efficiency gap (0.0010 versus Polymarket's 0.0017) enables superior monetary signal extraction. The cross-platform ensemble recovers the baseline Sortino ratio (from 0.330 to 0.362) while maintaining maximum drawdown invariance (-33.72%). To support this, we document the Phase 6 build-out: a multi-source, promotion-gated API featuring a canonical `SignalContract` schema. By eliminating implicit implementation discretion—specifically a hidden dictionary-ordering dependency in family selection—the system now achieves deterministic, auditable signal governance. The resulting `v0.6.4-market-ready` architecture provides an institutional-grade data plane that strictly separates public signal consumption from operator governance.

Keywords: prediction markets, signal infrastructure, ensemble methods, Becker efficiency gaps, API governance, market microstructure, CFTC DCMs
JEL Classification: G11, G17, C53, D84, C88

Introduction and Motivation

The first two Cassandra-Risk papers asked whether event-implied fragility could be transformed into a usable risk overlay. Paper 3 asks whether that overlay can be operationalized as a governed signal service across multiple platforms without losing the epistemic discipline that made the research credible.

A backtest can tolerate manual curation and implicit data conventions; a live signal API cannot. The moment a predictive framework is exposed as an enterprise service, every hidden assumption becomes a governance risk. Furthermore, relying on a single prediction market introduces platform-specific microstructure risks.

This paper introduces a cross-platform ensemble methodology to harvest microstructure errors via platform independence, and documents the canonical software architecture required to ingest, normalize, govern, and deliver these probabilities in a product-safe format.

Platform Microstructure and Ensemble Results

When structurally independent platforms price identical events, their ensemble should theoretically outperform single-source baselines by smoothing idiosyncratic liquidity shocks. To test this, we analyzed overlapping monetary policy contracts (e.g., FOMC rate decisions and emergency cuts) across Polymarket and Kalshi for the 2025–2026 period.

Microstructure Efficiency Gaps

Market efficiency was quantified using the Becker monetary gap:

- **Kalshi:** 72.1M trades, $18.26B Volume, Monetary Gap of 0.0010 (CFTC DCM Status)
- **Polymarket:** 32M+ trades, $10B+ Volume, Monetary Gap of 0.0017 (CFTC Status Pending)

Kalshi's 40% tighter monetary gap suggests cleaner policy extraction, while Polymarket provides crucial volume during thin order books.

Monetary Ensemble Weighting

Rather than raw concatenation, the ensemble applies a microstructure-aware weighting penalty based on the comparative efficiency gaps:

wkalshi=1−gapkalshigappolyw_{kalshi} = 1 - \frac{gap_{kalshi}}{gap_{poly}}wkalshi=1−gappolygapkalshi

This formula yields an approximate 65% weight to Kalshi signals for overlapping monetary events.

Performance Impact

The weighted ensemble configuration demonstrates quantifiable error reduction:

- **Polymarket-only (v0.6.3):** Sortino 0.330, CAGR 7.24%, Max Drawdown -33.72%
- **Raw Concatenation (Equal Weight):** Sortino 0.347, CAGR 7.81%, Max Drawdown -33.72%

- **Weighted Ensemble (Kalshi 65%):** Sortino 0.362, CAGR 8.12%, Max Drawdown -33.72%

The maximum drawdown invariance across 12 architectural variants from Papers 1 through 3 confirms the structural robustness of the underlying RSI engine.

Geopolitical Boundary Conditions

While monetary policy benefits from ensembles, geopolitical events exhibit severe heterogeneity. Applying a flat-theme Becker correction (0.0732) overcorrects heterogeneous events (e.g., ceasefires versus escalations). Cross-platform ensembling for these events currently adds noise rather than alpha, confirming the boundary limits established in Paper 2. Event-type stratification is required before geopolitical cross-platform benefits emerge.

System Design and Architectural Boundaries

To serve multi-platform ensembles safely, the API pipeline required a typed schema to replace loosely coupled payload dictionaries. Phase 6 introduces the `SignalContract` architecture.

The Canonical Schema

The core invariant of the v0.6.4 architecture is that platform provenance is erased at the `SignalContract` boundary. The RSI engine must consume only one canonical object; everything upstream is adapter logic, and everything downstream is governed aggregation. The schema explicitly enforces identity, probability bounds, temporal fields, and, crucially, an `aggregation_policy`.

Resolving Hidden Implementation Dependencies

During the schema migration, a regression test revealed that the live RSI moved sharply. The root cause was a hidden governance dependency: family representative selection had previously relied on implicit Python dictionary insertion ordering. A geopolitical cluster shifted from a cold contract to a hot contract simply because of payload ordering.

This was remediated by backfilling the `aggregation_policy` across all 54 governed registry rows. Family representation is now deterministic: `weighted_average` families anchor on the highest volume, while `max` families anchor on the highest calibrated probability.

Governance and Honest Emptiness

Discovered contracts do not silently enter the live RSI. They pass through a strict promotion workflow that wraps a `SignalContract` inside a `PromotionCandidate` for operator review. Approved contracts are written to a single machine-readable governed ledger (`signal_registry.json`) accompanied by a full decision audit trail.

Notably, during isolated testing, the live RSI returned exactly `1.0000` when no promoted governed signals were active. This "honest emptiness" proves the governance barrier functions correctly: discovery exists, but nothing crosses into the live signal without explicit operator admission.

Institutional API Interface

The final contribution of the v0.6.4 build is the explicit separation of public consumption and operator governance. Protected by API-key authentication and in-memory rate limiting (yielding HTTP 429 statuses when breached), the system exposes a hardened Tier 1 API surface.

As of March 26, 2026, the live governed ensemble outputs the following via `GET /v1/rsi/latest`:


```
{"rsi": 0.1006928493, "hazard": 8.93, "signals": 3, "updated": "2026-03-26T12:00Z"}
```


The active governed signals driving this output include:

- Polymarket 1551490 (Israel/Iran Escalation): Probability 0.86, Calibration: None
- Polymarket 616903 (Fed Emergency Rate Cut): Probability 0.26, Calibration: Becker
- Polymarket 561829 (Russia/Ukraine Ceasefire): Probability 0.005, Calibration: None

With 102 passing API contract tests, the system is commercially ready for developer deployment.

Conclusion and Future Work

Paper 3 proves that the Cassandra-Risk overlay can survive operationalization. Cross-platform ensembles deliver a 3.6% Sortino uplift via microstructure error harvesting, while the `SignalContract` schema and governed promotion workflow provide institutional-grade auditability. Cassandra is no longer solely a research architecture; it is a deterministic, typed, and bounded signal service.

Future work in Paper 4 will refine overlap-aware Kelly-optimal source weighting, complete the Kalshi RSA-PSS market ingest, and transition the data delivery layer to real-time WebSockets.

References

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